NON-MEASUREMENT SENSITIVE

MIL-D-28000
AMENDMENT 1
20 December 1988

MILITARY SPECIFICATION

DIGITAL REPRESENTATION FOR COMMUNICATION OF PRODUCT DATA: IGES APPLICATION SUBSETS

TO ALL HOLDERS OF MIL-D-28000

1. Make the following pen and ink changes:

PAGE 1

Paragraph 1.2, add the class: "Class IV - Geometry for NC Manufacturing"

PAGE 2

Change citation of Recommended Practices document to read:
"IGES Recommended Practices Guide - Version 4.0, March 1988."

PAGE 4

Paragraph 3.2, change last sentence to read:

"Unless otherwise noted, all references to entity types, form numbers, and data fields are to ANSI Y14.26M. Several specific extensions are identified which make use of IGES Version 4.0 entity constructs. These are noted by asterisks on the tables of entities for each class."

AMSC N/A AREA ILSS

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PAGE 24

Paragraph 3.2.3.5.2, Add to line for Field 24; "Default to zero"

PAGE 31

Change title of paragraph 6.6.1 to read: "Acronyms and abbreviations used ..."
In paragraph 6.6.1, insert new list item e and renumber list.
"e. NC Numerical Control"

PAGE 33

Change project number reference to: (Project ILSS - 0025)

The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement page	Page replaced
5-10	5-10
11	Reprinted without change
12-18	12-18
21	21
22	22
23, 23i and 23ii	23
24-26	24-26
26i-26viii	New Pages
27	Reprinted without change
28	28
29, 29i and 29ii	29
30	30

The margins of this amendment are marked with vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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Custodians:
Army - CR
Navy - SH
Air Force - 24
DLA - DH

Review activities:
Army - AM
Air Force - 01,02
NSA - NS
DCA - DC
NASA - NA
Others - NIST, DOE, GPO, NCS

User activities:
OSD - IR
Army - AL, AT, AV, EA, ER, GL, ME, MI, MR, SM, TE, TM
Navy - AS, EC, OS, SA, YD
Air Force - 11, 13, 14, 17, 18, 19, 68, 79, 99

Preparing Activity
OSD-CL
(Project ILSS - 0037)

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3.2.1. Class I Subset - technical illustrations. The Technical illustration subset of ANSI Y14.26M addresses entities that support the exchange of figures and illustrations normally found in a technical publication. In this application, emphasis is on visual clarity of figures and illustrations designed for human interpretation. The IGES Technical Illustrations Applications Guide provides additional guidance in this usage of IGES.

3.2.1.1. Information requirements and data functionality. Two dimension geometry and text annotation form the majority of data items although some non-geometric information is required as well. Information requirements for this application subset include:

a. Simple geometry in the form of points, lines, circular arcs, conic arcs and smooth curves.

b. Non-geometric attributes of line width and line font.

c. Text and annotation.

d. Data relationships, including the concept of subfigures.

3.2.1.2. ANSI Y14.26M entity subset specification. Table I lists the entities of this subset. Only ANSI Y14.26M entities which are enumerated in this table shall be used for representing technical publication illustration product definition data. Other valid ANSI Y14.26M entity types may be present in the file as described by 3.1.1 Additional requirements are placed on the Global Section of a file, and certain field value restrictions are also placed on the range of parameter values in both the Directory Entry (DE) and the Parameter Data (PD) Sections of a valid ANSI Y14.26M file.

3.2.1.3. File construction.

3.2.1.3.1. Start section. The following information shall be placed in the Start Section of the file:

a. Statement of conformance to this application subset and date.

b. Illustration number or identifier.

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TABLE I. - ANSI Y14.26M entity content of technical publication illustrations subset.

	ENTITY TYPE	FORM	ENTITY NAME	NOTES DE	NOTES PD
	100		Circular Arc	3	2
	102		Composite Curve	1	4
	104		Conic Arc - General	3	2
	106	11	Linear Planar Curve	1	2,5
	106	63	Simple Closed Area	1	2,5
	110		Line	1	2
	112		Parametric Spline Curve	1	2,6
	124	0	Transformation Matrix	7	8
	126		Rational B-Spline Curve	1	9
	212		General Note	11	2,12
	230		Sectioned Area	1	2
|	308		Subfigure Definition	7,13	4
|	404		Drawing		10
|	406	16	Drawing Size Property	14	15
| *	406	18	Intercharacter Spacing	1	16
	408		Subfigure Instance	3	2
	410		View	7	17
	412		Rect Array Sub Instance	3	2
	414		Circ Array Sub Instance	3	2

NOTES FOR TABLE I:

\| * 1. DE Field 4, Line Font Pattern, shall be 1, 2, 3, 4 or 5.
DE Field 5, Level, shall be 0
DE Field 6, View Pointer, shall be 0
DE Field 7, Transformation Matrix Pointer, shall be 0
DE Field 8, Label Display Pointer, shall be 0

\| 2. PD Index Values for Z coordinates shall be 0.0

\| * 3. DE Field 4, Line Font Pattern, shall be 1, 2, 3, 4 or 5.
DE Field 5, Level, shall be 0
DE Field 6, View Pointer, shall be 0
\| DE Field 7, Transformation Matrix Pointer, shall allow Z axis
\| rotation only.
DE Field 8, Label Display Pointer, shall be 0

* Denotes an IGES Version 4.0 capability

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4. PD Index values shall point only to other entity types within this subset.

5. PD Index 1, Interpretation Flag, shall be 1

6. PD Index 3, Planar, shall be 2

7. DE Field 7, Transformation Matrix, shall be 0.

8. Translation and rotation are restricted to XY plane. PD Indices R13, R23, R31, R32 and T3 shall be 0.0, and R33 shall be 1.0 or -1.0

9. PD Index 3, PROP1 (Planar), shall be 1 ZK shall be 0.0
XNORM and YNORM shall be 0.0 ZNORM shall be 1.0

10. PD Index 1, Number of view pointers, shall be 1
PD Index 5, Number of annotation entities, shall be 0
PD Index 6, Number of associativity pointers, shall be 0
PD Index 7, Number of property pointers, shall be 1
PD Index 8 shall be a DE pointer to a Property (406, Form 16)

11. DE Field 5, Level, shall be 0
DE Field 6, View Pointer, shall be 0
DE Field 7, Transformation Matrix Pointer, shall be 0
DE Field 8, Label Display Pointer, shall be 0

* 12. PD Index 5, Font Characteristic, shall be 1, 1001 or 1002. If a pointer to a property entity (406, Form 18) is used to control intercharacter spacing, then any IGES Version 4.0 font value may be used.

13. DE Field 9b, Subordinate Switch, shall be 1.

14. DE Field 5, Level, shall be 0.
DE Field 9b, Subordinate Switch, shall be 2.

15. PD Index 1, Number of Property Values, shall be 2.
PD Index 2 shall be X size.
PD Index 3 shall be Y size.

16. PD Index 1, Number of Property Values, shall be 1.
PD Index 2 shall be the intercharacter spacing value.

17. PD Index 2, Scale, shall be 1.0.
PD Indices 3-8 shall be 0.

* Denotes an IGES Version 4.0 capability

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3.2.1.3.2. Global section. Fields in the global section shall be restricted to certain ranges and shall not be defaulted except as noted below for parameters 1, 2, 12 and 24.

Field	Value	Required	
1	,	N	Default to ,
2	;	N	Default to ;
3-6		Y	
7-11		N	
12		N	Default to Field 3
13	1.0	Y	
14	1,2,4-11	Y	
15		N	
16		Y	
17		Y	
18-20		Y	
21		N	
22		N	
23	4-6	Y	
24	0-7	N	Default to zero

3.2.1.3.3. Directory entry section. See Notes for table I for restrictions placed on the parameters in the directory entry section.

3.2.1.3.4. Parameter data section. See Notes for table I for restrictions placed on the parameters in the parameter data section.

3.2.1.4. Mapping of information content to ANSI Y14.26M subset entities. Illustration geometry shall be mapped into ANSI Y14.26M geometry entities and annotation entities. The composite curve, subfigure definition and subfigure instance entities shall be used to organize the illustration information to preserve any required data relationships. Line width and line font information shall be represented by the appropriate global and directory entry parameters.

Several entity structures in this subset have been included to keep the file size to acceptable levels. For instance, the use of subfigures greatly reduces file size where illustration details are repeated. Similarly, the general note entity is a compact method of representing text annotation as compared with the stroking of each character using line and arc geometry. However, there are instances where a system lacks the sophistication of subfigure entity constructs or it is desired to stroke the text for a special appearance not otherwise attainable. Use of new IGES Version 4.0 constructs, like the intercharacter spacing property for proportional text, is encouraged to reduce file lengths. (See 6.3.)

3.2.1.5. Data accuracy requirements. All data transformations shall maintain an accuracy of at least 0.001 units on all parametric and coordinate values and all measurable dimensions.

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3.2.1.6. User conventions and data organization. The Technical Illustration subset places requirements on the start section for two defined human-readable record types. A minimum complexity drawing view entity combination along with its drawing size property shall be used to assure an illustration being created on all receiving systems.

3.2.2. Class II Subset - engineering drawings. This engineering drawing subset of ANSI Y14.26M shall be used to encode product data being acquired in accordance with DOD-D-1000 for delivery in digital form. The data exchange shall preserve the requirements of style and content as set forth by DOD-STD-100 and DOD-D-1000. Exchange emphasis is on completeness, visual equivalency for human interpretation and functionality of the received drawing model.

| 3.2.2.1. Information requirements and data functionality. Currently, engineering drawings form the mainstay of product definition and form the accepted medium for viewing computer generated data. Geometry and text annotation form the majority of data items. Extensive use is made of dimensions, sectioned areas, text notes, and feature control symbols. Non-geometric information in the form of color, line width, line font, and level is needed as well. Finally, relationships among the data help to structure the large quantities of data. Information requirements for this application subset include:

a. Simple geometry in the form of points, lines, circular arcs, conic arcs, and smooth curves.

b. Non-geometric attributes of color, line width and line font.

c. Data relationships, including the concept of subfigures.

d. Data organization methods such as level.

e. Part name, drawing number, formal identification, and drawing revision.

f. Annotation represented by the corresponding ANSI Y14.26M entity type thus retaining its intended functionality.

3.2.2.2. ANSI Y14.26M entity subset specification. Table II lists the entities of this subset. Only ANSI Y14.26M entities which are enumerated in this table shall be used for representing engineering drawing product | definition data. Other valid ANSI Y14.26M entity types may be present in the | file as described by 3.1.1. Additional requirements are placed on the global | section of a file, and certain field value restrictions are placed on the | range of parameter values in both the directory entry and the parameter data | sections of a valid ANSI Y14.26M file.

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TABLE II. - ANSI Y14.26M entity content of engineering drawing subset

	ENTITY TYPE	FORM	ENTITY NAME	NOTES DE	NOTES PD
	100		Circular Arc	1	
	102		Composite Curve	1	2
	104		Conic Arc	1	5
	106	11	Linear Planar Curve	1	6
	106	12	Coordinate Triples	1	6
	106	20	Centerline Thru Points	1	10
	106	21	Centerline Thru Centers	1	10
	106	31	Section Form 31	1	10
	106	32	Section Form 32	1	10
|	106	33	Section Form 33	1	10
|	106	34	Section Form 34	1	10
|	106	35	Section Form 35	1	10
|	106	36	Section Form 36	1	10
|	106	37	Section Form 37	1	10
|	106	38	Section Form 38	1	10
	106	40	Witness Line	1	12
	106	63	Simple Closed Area	1	
|	* 108	0	Unbounded Plane	1	7
	108	1	Bounded Plane	1	16
	110		Line	1	
	112		Parametric Spline Curve	1	8
	114		Parametric Spline Surface	1	8
	116		Point	1	
	118		Ruled Surface	1	2
	120		Surface of Revolution	1	2
	122		Tabulated Cylinder	1	2
	124	0	Transformation Matrix	1	
	124	1	Transformation Matrix	1	
|	126	0	Rational B-Spline Curve	1	
	128	0	Rational B-Spline Surface	1	
	128	2	RBS Right Circular Cylinder	1	
	128	3	RBS Cone	1	
	128	4	RBS Sphere	1	
	128	5	RBS Torus	1	
	128	9	RBS General Quadratic	1	

\| * Denotes an IGES version 4.0 capability.

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TABLE II. (Continued) - ANSI Y14.26M entity content of engineering drawing subset

ENTITY	FORM	ENTITY NAME	NOTES DE	NOTES PD
130		Offset Curve	1	2
140		Offset Surface	1	2
142		Curve on a Parametric Surface	1	2
144		Trimmed Parametric Surface	1	2
202		Angular Dimension	1	
206		Diameter Dimension	1	
210		General Label	1	
212		General Note	1	3
214		Leader Arrow	1	9
216		Linear Dimension	1	
218		Ordinate Dimension	1	
220		Point Dimension	1	13
222		Radius Dimension	1	
228		General Symbol	1	2,11
230		Sectioned Area	1	2
304		Line Font Definition		
308		Subfigure Definition	1	2,14
314		Color Definition	1	
402	3	Views Visible	1	
402	4	Views Visible, Color, Line Font	1	
402	15	Ordered Group wo Back Pointer	1	
404		Drawing	1	4
406	1	Definition Levels	1	
406	3	Level Function	1	
406	5	Line Widening	1	15
406	15	Name		1
406	16	Drawing Size	1	
406	17	Drawing Units	1	
408		Subfigure Instance	1	
410		View		1

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NOTES FOR TABLE II:

1. DE Field 8, Label Display Pointer, shall be 0

2. PD Pointer values shall point only to other entity types within this subset.

* 3. Font Characteristic shall be 1, 1001, 1002 or 1003.

4. Drawing origin shall be the lower left-hand corner, no negative coordinates are allowed once all appropriate offsets and rotations are applied.

5. Conic coefficient B (PD Index 2) shall be zero (0). An associated matrix shall be used to rotate/translate the conic to its position in space.

6. N1 (the number of Points) shall be 3 or more.

7. Shall only appear in the file if it is used with an entity type 410 as a clipping plane.

8. CTYPE shall be 2 or 3.

9. Shall have at least one segment. First segment can be zero length. Neither arrowhead height nor width shall be zero except for Form 4 where both shall be zero or Form 9 where the height can be zero.

10. Shall have an even number of points.

11. Can be used only for annotation.

12. Shall contain at least two segments. Only first segment can be zero length.

13. Leader shall be Form 4.

14. DE Field 7 shall be 0.

15. Only allowed use of the line widening property shall be when physical model space width is important.

16. Shall always point to a boundary curve. Parameter fields 6-9 shall be ignored.

* Denotes an IGES version 4.0 capability.

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3.2.2.3. Entity construction. The following entities (in entity number order) have particular meanings when used for engineering drawings. The requirements in this section shall be met in all conforming data files and by all translator implementations of the Class II subset of ANSI Y14.26M.

102 Composite Curve. Composite curves are intended for showing connectivity and continuity. They shall contain a minimum of two entities one of which shall be a curve.

124 Transformation Matrix. Defining matrices (Forms 0 and 1) are used to position an entity into model space from its definition space. When entities share the same plane of definition, they shall use the same transformation matrix to avoid multiple identical matrices being included in the file. If an entity contains translation information in its PD section, a transformation matrix shall not be used to translate the entity.

202 Angular Dimension. This entity shall have a general note, two leaders, and a vertex point. The Z displacement of the vertex point can be attained from any of the subordinates.

206 Diameter Dimension. This entity shall have a general note, at least one leader, and an arc center (may not be defaulted). The Z displacement of the arc center can be attained from the subordinates. The arc center shall be valid. If multiple leaders occur, the first segment of each leader shall be co-linear and opposite in direction.

210 General Label. This entity shall have at least one leader and one general note.

212 General Note. General notes shall use a font code to minimize the number of text strings in the note. At least one string is required, but the number of different strings shall be minimized. Form numbers plus position information on each string shall be used. The Rotation Angle field shall contain the string angle. Transformation matrices shall not be used for string angles. Null strings are allowed and may be used to pattern a note into one of the standard forms.

216 Linear Dimension. This entity shall have two leaders and at least one general note. The witness lines shall be parallel.

218 Ordinate Dimension. This entity shall have a general note and either a leader or witness line. This entity shall not be used in place of the general label. The leader shall contain only one segment.

222 Radius Dimension. This entity shall have a general note, at least one leader, and an arc center (may not be defaulted). Z displacement of the arc center can be attained from any of its subordinates. Arc center shall be valid. If multiple leaders occur, the first segment of each

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shall be colinear. If two leaders are used, one and only one shall be of Form 4.

228 General Symbol. This entity shall be used for annotation. Flags in the subordinate entities (even geometry) shall be set to annotation. This entity shall not be used in place of a subfigure or group. The entity shall have a general note and at least one piece of geometry. All subordinate entities to the general symbol entity shall be co-planar.

230 Sectioned Area. This entity is used to provide for area fills. The normal distance between lines is constant and shall not be zero. Wider spaces between lines are to be considered blank lines with regard to normal distances. Boundary curve and islands are closed and not self-intersecting. Island interiors are mutually disjoint. Islands lie in the interior of the boundary curve. Boundary curve and islands can be logically dependent and do not need to be marked as annotation if they are not annotation.

404 Drawing. The drawing entity defines the basic engineering drawing format for each sheet. One drawing entity shall exist in the file for each sheet of an engineering drawing. All entities pointed to by the drawing entity shall be physically dependent and shall be flagged as annotation.

406 Form 15 Name. This entity is used to convey the drawing name. A Drawing Name using Entity 406, Form 15 is required. If there is no name available in the architecture of the system, the pre-processor shall insert one, a reasonable default being global parameter 3.

406 Form 5 Line Widening. The only allowed use of the line widening property is the case when physical model space width is important. It shall not be used to indicate plotter width or line weight.

406 Form 16 Drawing Size. A drawing size property shall be included for each drawing entity in the file. If drawing size is not in the architecture of the system, the pre-processor shall insert one.

406 Form 17 Drawing Units. A drawing units property shall be included for each drawing entity in the file. If drawing units is not in the architecture of the system, the pre-processor shall use the defaults given in Global Parameters 14 and 15.

In general, the simplest entity type shall be used to represent each piece of geometry, and zero size entities shall not be used. For instance, a B-Spline Curve shall not be used to represent a circular arc, a zero length line entity or a zero diameter circle shall not occur in the file.

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All 200 series entities shall be flagged as annotation and shall be parallel to the viewing plane, except for crosshatching. All dependent entities with parent 200 entities shall be flagged as annotation and shall be coplanar with their parent. The only exception is for geometry entities associated with the Sectioned Area (230 and 106 Forms 31-38). Any annotation entity which points to multiple leaders and multiple witness lines shall order those leaders and witness lines so that leader-1 corresponds logically to witness-1 and leader-2 to witness-2.

3.2.2.4. File construction.

3.2.2.4.1. Start section. The following information shall be given in the Start Section of the file:

a. Statement of conformance to this application subset and date.

b. All part and drawing identification required in the drawing title block by DOD-STD-100 and DOD-D-1000.

c. Revision letters of most recent change to each drawing sheet in the file.

d. Performing organization, date and contract number.

e. Intended drawing size letter and number of drawing sheets in this file.

f. Data organization method with contents of each level.

3.2.2.4.2. Global section. Fields in the global section shall be restricted to certain ranges and shall not be defaulted except as noted below for parameters 1, 2, 12 and 24.

Field	Value	Required	
1	,	N	Default to ,
2	;	N	Default to ;
3-11		Y	
12		N	Default to Field 3
13	1.0	Y	
14	1-11	Y	
15		Y	
16		Y	
17		Y	
18-22		Y	
23	4-6	Y	
24	0-7	N	Default to zero

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3.2.2.4.3. Directory entry section. See notes for table II for restrictions placed on the parameters in the directory entry section. In addition, the following capabilities shall be provided and shall be supported for all entities:

a. Line font pattern.
b. The level number field shall be zero or positive except where necessary to maintain the meaning of the referenced entity.
c. View pointer.
d. Translation matrix.
e. Blank status flag.
f. Subordinate entity switch.
g. Entity use flag.
h. Hierarchy status flag.
i. Line weight number.
j. Color number.
k. Form number.

3.2.2.4.4. Parameter data section. See notes for table II for restrictions placed on the parameters in the parameter data section.

3.2.2.5. Mapping of information content to ANSI Y14.26M subset entities. Engineering drawing geometry shall be mapped into ANSI Y14.26M geometry entities and organized as necessary with the composite curve, subfigure definition, and subfigure instance entities. Text shall be represented by the general note entities and shall not be represented as geometry. Annotation including dimensions, general symbols, and centerlines shall be represented by their named ANSI Y14.26M entity and shall not be represented as geometry. Use of IGES Version 4.0 constructs such as the new dotted line font (DE Field 4=5) is encouraged to reduce large file lengths. (See 6.4.)

3.2.2.6. Data accuracy requirements. All data transformations shall maintain an accuracy of 1.0×10^{-6} units on all parametric and coordinate values and all measurable dimensions.

3.2.2.7. User conventions and data organization. A minimum complexity drawing and view entity combination shall be used to assure a part model being created on all destination systems. A drawing size property shall be used to define drawing limits. As specified in the contract or other form of agreement, data shall be organized as one drawing per file with multiple sheets permitted, or shall be restricted to one sheet per file.

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3.2.3. Class III Subset - electrical/electronic applications. The electrical/electronic applications subset of ANSI Y14.26M defined herein addresses the representation and exchange of data for electrical and electronic products including printed wiring boards, printed wiring assemblies, hybrid microassemblies, flexible printed wiring harnesses, cables, and conventional wiring harnesses. Emphasis is on component and circuit element descriptions, their placement, their connectivity, and the routing of electrical paths. This subset supports both the physical view of the product and the logical view of the product. The physical view includes product representations such as the assembly placement, the etch artwork, pad layouts, etc., while the logical view of the product includes representations such as the netlist, schematic, etc. Production of drawings and direct extraction of information for subsequent computerized process steps are supported. Inclusion of both the logical view information and the physical view information in the same file is not precluded. Completeness and functionality requirements of the received model for design, manufacturing and testing purposes are the basis of this subset. The IGES Electrical Applications Guide provides additional guidance in this usage.

3.2.3.1. Information requirements and data functionality. ANSI Y14.26M files for electronic products rely heavily on both geometry and logical relationships. Components or circuit elements are defined to include graphic outline and appearance features, logical references to properties, logical ownership of connection points, and logical relations to other components or circuit elements. The relationships include connectivity between components or circuit elements, connecting geometry, signal information, and property and file data structure relationships. Examples include properties (component or circuit element value, drill-hole size, line trace thickness), attributes (current, resistance, wattage), transformation matrices, text display templates and simple, predefined display symbols.

Three dimensional geometry and text annotation form the majority of data items. Non-geometric information in the form of color, line width, line font, and level is required as well. Finally, relationships among the data such as subfigures and connectivity are required to help structure the large quantities of data.

Information requirements for this application subset include (but are not limited to):

a. Component or circuit element descriptions of three dimensional geometry in the form of points, lines, circular arcs and curves.

b. Non-geometric attributes of color, line width and line font.

c. Electrical/electronic attributes of current, resistance, and wattage.

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d. Data relationships including the concept of subfigures, network subfigures, external references, connect points, connectivity and signals.

e. Data organization methods such as Level.

f. Part name, drawing number, formal identification and change history.

g. Text annotation for display.

h. Drafting entities such as dimensions and text notes.

i. Printed wiring board layering for interconnection routing, power planes and ground planes.

j. Back annotation using a file containing only the changes to the netlist.

3.2.3.2. Connectivity. Forming a logical connection between two or more items requires the ability to represent the following:

- The logical connection between a unique network subfigure (e.g. component) and each of its logical points of connection.

- The signal and its identification (if any) at the components unique connection point.

Forming a physical connection between two or more items requires the ability to represent the following:

- The exact location of each connection point.

- The ability to represent a physical connection between the two items (whether by path, via, or wire).

The term "connect node" refers to a data base entity which represents the logical, and optionally the physical, location of connection. The term "link" refers to the logical representation of the signal formed, and "signal name" refers to the signal identifier. The term "join" refers to the data base entity or entities which represent the physical connection between the items.

Each item to be connected requires a connect point to represent each possible connection point of the item. A signal may be formed between any such items by a link which references the connect points to be connected. This creates an associativity between the connect points, and thus the connected items. The signal name may be used to uniquely identify the particular signal formed. The join may be used to provide a graphical representation of the signal. In electrical applications, the join is most often represented by a line (schematic) or a widened line (printed wiring board).

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TABLE III - (Continued) ANSI Y14.26M entity content of electrical/electronic applications subset

	ENTITY	FORM	ENTITY NAME	NOTES DE	NOTES PD
	312	0	Text Display Template - Abs		
	312	1	Text Display Template - Incr		
	320		Network Subfigure Definition		
*	322		Attribute Table Definition		
	402	1	Group w BP Instance		
	402	7	Group wo BP Instance		
	402	18	Flow Associativity		3
	404		Drawing		
	406	1	Definition Levels Property		
	406	2	Region Restriction Property		
	406	3	Level Function Property		
	406	5	Line Widening Property		
	406	6	Drilled Hole Property		
	406	15	Name		
	406	16	Drawing Size		
	408		Subfigure Instance		
	410		View		
	412		Rect Array Sub Instance		
	414		Circ Array Sub Instance		
	416		External Reference		
	420		Network Subfigure Instance		
*	422		Attribute Table Instance		

NOTES FOR TABLE III:

* 1. DE Field 4, Line Font Pattern, shall be 1, 2, 3, 4 or 5
DE Field 5, Level, shall be 0 or positive
DE Field 6, View Pointer, shall be 0
DE Field 8, Label Display Pointer, shall be 0
DE Field 13, Color Number, shall be 0-8

* 2. Font Characteristic shall be 1, 1001, 1002 or 1003.

3. PD Type Flag shall be set to 1 or 2 when both logical and physical product definitions coexist in the same file. Function Flag shall be set to 1

4. ZT shall be set to 0.0

* Denotes an IGES version 4.0 capability.

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3.2.3.4. Entity construction. The following entities (in entity number order) have particular meanings when used for electrical product data. The specifications in this section shall be used in all conforming data files and by all translator implementations of the Class III subset of ANSI Y14.26M.

100 Circular Arc Entity. The electrical use of this entity is in the geometric representation of component parts and their symbolic representations. In such usage, it is usually part of a subfigure. It may also be used as connection geometry.

102 Composite Curve Entity. The electrical use of this entity is in the geometric representation of component parts and their symbolic representations. In such usage, it is usually part of a subfigure. It may be used as connection geometry.

104 Conic Arc Entity. The electrical use of this entity is in the geometric representation of component parts and their symbolic representations. In such usage, it is usually part of a subfigure. It may be used as connection geometry.

106 Copious Data Entity. Forms 11 and 12 of this entity may be used as connection geometry. Forms 20, 21, and 40 are used in the drafting documentation of the product. Form 63, Simple Closed Area, may be used to define an auto-router restriction area or a PC-defined (or IC-defined) area having special attributes.

108 Plane Entity. Certain layers of PC design are designated as ground, power, or heat sink, and as such are large conductive areas. These layers, as well as larger curved or rounded conductive areas on other layers, are best defined by the Plane entity. Note that the form number indicates whether the bounded region is positive or void (i.e., copper-clad area or cutout).

110 Line Entity. The Line entity has several important uses in the electrical application. It may be used to construct component outlines, or to represent connection geometry.

112 Parametric Spline Curve Entity. The electrical use of this entity is in the geometric representation of component parts and their symbolic representations. In such usage, it is usually part of a subfigure. It may be used as connection geometry.

116 Point Entity. The point entity is used to locate features that do not participate in connectivity, (e.g., a mounting hole)

124 Transformation Matrix Entity. Electrically, a transformation matrix entity may be used to rotate subfigures to other than normal (top up) positions or may be used to reverse the direction of an arc draw (determinant is less than zero). Generally, rotations are about the Z

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axis for PC and IC constructs, but may be about any axis for 3D cabling files.

125 Flash Entity. The Flash entity may be used to represent a repetitive artwork feature which is usually produced by photo-optical means. Examples include PC pads, clearances, and hybrid and IC features.

126 Rational B-Spline Curve Entity. The electrical use of this entity is in the geometric representation of component parts and their symbolic representations. In such usage, it is usually part of a subfigure. It may be used as connection geometry.

132 Connect Point Entity. The Connect Point entity is used to represent a point of connection. A subfigure defining an electrical component typically uses the Connect Point entity to represent a pin of the logical or physical component or symbol. A Connect Point may also be used in a stand-alone mode, representing a via hole for example. The drilled hole property may be attached to the connect point if appropriate.

212 General Note Entity. A General Note is used to display constant text. Design notes would require a General Note, for example.

230 Sectioned Area Entity. The electrical use of this entity is in the geometric representation of large plated or unplated board areas (e.g., power and ground planes or keep-out areas). It may be used as connection geometry.

302 Associativity Definition Entity. When the originating system provides for a relationship not included among the ANSI Y14.26M predefined associativities, this entity is required. Possible uses are to relate subfigures to entities in other data bases (e.g., circuit analysis or text requirements) or for back-annotation purposes.

308 Singular Subfigure Definition Entity. This entity may exist in a library of physical or logical primitive parts in the originating system. This entity shall not be used for components which participate in connectivity.

312 Text Display Template Entity (Form 0: absolute; Form 1: incremental). The Text Display Template may be used to display text which may be unique in each instance of the defined entity, (A pin number, for example).

320 Network Subfigure Definition Entity. For PC, cable, IC, hybrid and gate array usage, a subfigure usually represents a component and its required contacts. This entity is normally a library physical or logical figure in the originating system.

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322 Attribute Table Definition Entity. This entity provides electrical and electronic attributes which apply to the components and circuit elements.

402 Associativity Instance Entity. This entity relates other entities within a file to provide a set with a common meaning. Those associativities which are predefined by ANSI Y14.26M are identified by ANSI Y14.26M form numbers (e.g., form 18: Flow). The user-defined associativities are defined by an entity 302 and its form number.

406 Property Entity. The use of a property to indicate the meaning or purpose of a geometric entity applies to electrical constructs as well as general graphics. A Connect Point entity may point to the Drilled Hole property. A Plane entity or Simple Closed Area entity may point to the Region Restriction property. Any property, however, may point to the Text Display Template, with the text string specified in the property. In this case, the Text Display Template locates the text display.

408 Singular Subfigure Instance Entity. This entity allows a non-electrical primitive component to be instanced in a number of unique locations. This entity shall not be used for components which participate in connectivity.

412 Rectangular Array Subfigure Instance Entity. This entity may be used to instance multiple subfigures, but shall not be used for instancing connectivity related entities (e.g., Connect Point entity (132), Network Subfigure Instance entity (420)).

414 Circular Array Subfigure Instance Entity. This entity may be used to instance multiple subfigures, but shall not be used for instancing connectivity related entities (e.g., Connect Point entity (132), Network Subfigure Instance entity (420)).

416 External Reference Entity. This entity provides a link between an entity in a referencing file and the definition or a logically related entity in a referenced file. The capability is essential to the concept of component libraries.

420 Network Subfigure Instance Entity. This entity allows an electrical component to be instanced in a number of unique locations. Note that owned Connect Point entities shall be instanced with this entity.

Several of the geometry entities may be used to implement physical connections on schematics or physical products such as printed circuit boards, hybrids, integrated circuits, and flex cables. The use of the entity may be defined by the level function property or DE level field. When the geometry is used for display purposes, its width will usually be determined by DE field 12. When the geometry is used for physical implementation, its width will usually be determined by use of the Line Widening property (406 form 5).

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3.2.3.5. File construction.

3.2.3.5.1. Start section. The following information shall be given in the Start Section of the file:

a. Statement of conformance to this application subset and date.

b. All part and drawing identification required in the drawing title block by DOD-STD-100 and DOD-D-1000.

c. Revision level of the file.

d. Performing organization, date, and contract number.

e. Data organization method with contents of each level.

3.2.3.5.2. Global section. Fields in the global section shall be restricted to certain ranges and shall not be defaulted except as noted below for parameters 12, 17 and 24.

Field	Value	Required	
1	,	Y	
2	;	Y	
3-11		Y	
12		N	Default to Field 3
13	1.0	Y	
14	1-11	Y	
15		Y	
16		Y	
17		N	
18-22		Y	
23	4-6	Y	
24	0-7	N	Default to zero

3.2.3.5.3. Parameter data section. See Notes for Table III for restrictions placed on the parameters in the parameter data section.

3.2.3.5.4. Directory entry section. See Notes for Table III for restrictions placed on the parameters in the directory entry section. In addition, the following capabilities shall be provided and shall be supported for all entities:

a. Level Number Field. All values shall be positive except where necessary to maintain the meaning of the referenced entity.
b. Subordinate Entity Switch.
c. Entity Use Flag.
d. Hierarchy Status Flag.
e. Blank Status Flag.

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3.2.3.6. Mapping of information content to subset entities. A connect node is represented by the connect point entity while a text node is represented by the text display template entity. The flow associativity entity is used to represent a signal and contains the link, signal name, and pointers to the join entities. The network subfigure entities (definition and instance) represent electrical components which participate in signals. A number of property entities are also used, as explained below.

3.2.3.6.1. Network subfigure construction. A component is constructed using a network subfigure definition entity. The graphics represent the component geometry the same as referenced by the singular subfigure definition entity. A separate set of pointers is provided which points to the defining connect point entities. These connect point entities define the locations and characteristics of the component's pins. Properties (for example the Part Name property) may be attached to the network subfigure definition entity.

3.2.3.6.2. Connect points. A component pin (or surface mounted device pad, IC I/O port, lead frame, schematic symbol lead, etc.) is represented by a connect point entity. The connect point entity is used in both logical and physical product designs. The physical location may be specified, along with several characteristic flags (connection type, function type, I/O direction). There is a pointer to the parent network subfigure entity (definition or instance) which provides a much needed association for signal processing. An additional subfigure instance pointer is provided for connect point display. This allows a graphical symbol to be displayed, representing the connect point. The pin number is provided in the function connect point identifier field along with a pointer to a text display template for pin number display. A pin function name is provided in the connect point function name field along with a pointer to a text display template for its display.

3.2.3.6.3. Signal construction. A signal representing one set of electrically common connect points is constructed using the flow associativity entity. It contains pointers to other associated flow associativity entities, the connect point entities participating in the signal (this is the Link), and the join entities representing the geometry of the signal (logical or physical). Also contained is a list of signal names which may be used to identify the signal, along with a set of pointers to text display template entities which may be used to display the first signal name in a number of locations. Two characteristic flags determine the signal type (logical or physical), and the function type (fluid flow or electrical signal).

A signal, then, is represented by one flow associativity entity pointing to a set of electrically common connect points. This is the link. The join entities represent the physical display geometry of the signal. For a schematic (logical), a line without width is typically used. For a printed board (physical), geometry with the line widening property is typically used. A number of signal names may be associated with the signal. Multiple displays of the first, or primary, name are possible.

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The components participating in a signal are represented by the network subfigure instance entity. Note that the connect point entities belonging to a component are instanced along with the subfigure. This is necessary to allow a subfigure to participate in a number of different signals while retaining unique component/pin identification. Each component is usually identified by a reference designator, supplied by the Primary Reference Designator field.

3.2.3.6.4. Information display. Throughout this discussion references have been made to the text display template entity. This entity allows text embedded in an entity to be displayed without the redundant specification of the text string. There are two reasons for this feature. First, it eliminates a possible source of error by allowing the information to be specified in only one place. Second, it reduces the file size overhead. This entity exists in two forms, absolute and incremental. The absolute form provides an exact location for display of the information. The incremental form provides an offset to be applied to the parent entity's location which provides the exact location for display of information such as pin numbers. When a direct pointer for information display is provided, the base location is readily determined from the parent entity's location, such as when displaying a pin number. In the case of property value display, the base location shall be determined by remembering the location of the property entity's parent entity. This would occur when displaying the part name. Also, in this case, the pointer to the text display template entity is located in the additional pointers section of the property entity parameters.

3.2.3.6.5. Attributes. Each electronic device has various properties, or attributes, which describe the behavior and characteristics of the device. These attributes must also be represented in the entities of ANSI Y14.26M. The Attribute Table Definition and Instance entities are used to do this. These entities use one extensible format which can hold a variety of attribute definitions, either singly or in combination. By defining the various attribute combinations once, they can be referred to by all entities which have those attributes.

3.2.3.6.6. Additional considerations. The situation is exactly the same for both logical and physical representations. The only differences arise in the subfigure and join entities used. In fact, an ANSI Y14.26M file may contain representations for both the schematic and the board. The flow associativity entity contains a type flag to indicate the connection type (logical or physical). In this case, one flow associativity entity would represent the logical connection and a second flow associativity entity would represent the physical connection. The two associativities would be related by the pointers provided in the flow associativity.

3.2.3.7. Data accuracy requirements. All data transformations shall maintain an accuracy of 1.0×10^{-6} units on all parametric and coordinate values and all measurable dimensions.

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3.2.3.8. User conventions and data organization. None

3.2.4. Class IV Subset - Geometry for NC Manufacturing. This numerical control machining subset of ANSI Y14.26M shall be used to encode product data for the subsequent purposes of manufacturing by numerical control. As such it is designed to directly support the geometry data needs of process planning and numerical control cutter path generation. The data exchange shall preserve the precision and accuracy of all wireframe and surface geometry as well as the first order continuity between geometry entities. Exchange emphasis is on completeness and functionality of the received part model.

3.2.4.1. Information requirements and data functionality. Geometry and text annotation form the majority of data items. Geometry data is used to describe the nominal shape of the product either as a 2-D wireframe or as a 3-D wireframe model with surfaces. 2-D descriptions are used to describe the profile of turned parts or of 2-D parts such as found in sheet metal work. 3-D descriptions are used for multiaxis machining.

Nominal dimensions given in text annotation shall agree with the corresponding values in the geometry data description. A major purpose of text annotation shall be to represent tolerances on the geometry. Annotation is also used to describe material specifications and administrative data. Non-geometric information in the form of color, line width, line font, and level is needed as well. Finally, relationships among the data are defined to help structure the large quantities of product data.

3.2.4.2. ANSI Y14.26M entity subset specification. Table IV lists the entities of this subset. Only ANSI Y14.26M entities which are enumerated in this table shall be used for representing geometry for NC manufacturing. Other valid ANSI Y14.26M entity types may be present in the file as described by 3.1.1. Additional requirements are placed on the global section of a file, and certain field value restrictions are placed on the range of parameter values in both the directory entry and the parameter data sections of a valid ANSI Y14.26M file.

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TABLE IV. - ANSI Y14.26M entity content of geometry for NC manufacturing subset

ENTITY	FORM	ENTITY NAME	NOTES DE	NOTES PD
100		Circular Arc	1	17
102		Composite Curve	1	17
104		Conic Arc	1	2,17
106	1	Coordinate Pairs	1	3
106	2	Coordinate Triple	1	4,17
106	11	Linear Planar Curve	1	5,17
106	12	Coordinate Triples	1	6,17
106	20	Centerline Thru Points	1	
106	21	Centerline Thru Centers	1	
106	40	Witness Line	1	
106	63	Simple Closed Area	1	
108		Plane	1	
110		Line	1	17
116		Point	1	7,17
118	1	Ruled Surface	1	17
120		Surface of Revolution	1	17
122		Tabulated Cylinder	1	17
124		Transformation Matrix	1,8	
126		Rational B-Spline Curve	1,9	17
128		Rational B-Spline Surface	1,10	17
202		Angular Dimension	1	
206		Diameter Dimension	1	
210		General Label	1	11
212		General Note	1	12
214	1	Leader Arrow, Wedge	1,13	14
214	2	Leader Arrow, Triangle	1	
214	3	Leader Arrow, Filled Triangle	1	
214	11	Leader Arrow, Open Triangle	1	
216		Linear Dimension	1	
218		Ordinate Dimension	1	
220		Point Dimension	1	
222		Radius Dimension	1	
402	3	Associativity, Views Visible		
402	4	Associativity, VV, Color, Line Font		
402	7	Associativity, Group w/o Back Pointers	1,15	
402	9	Associativity, Single Parent	1	16
404		Drawing		
406	16	Drawing Size Property		
406	17	Drawing Units Property		
410		View		

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NOTES FOR TABLE IV:

* 1. DE Field 4, Line Font Pattern, shall be 1, 2, 3, 4 or 5
DE Field 9c, Entity Use, shall be 00, 01, 02, or 03
DE Field 9d, Hierarchy, shall be 01

2. The conic shall be defined in the standard position.

3. PD Index 1, Interpretation Flag, shall be 1
PD Index 2, Number of n-tuples, shall be greater than 1

4. PD Index 1, Interpretation Flag, shall be 2
PD Index 2, Number of n-tuples, shall be greater than 1

5. PD Index 1, Interpretation Flag, shall be 1
PD Index 2, Number of n-tuples, shall be greater than 2

6. PD Index 1, Interpretation Flag, shall be 2
PD Index 2, Number of n-tuples, shall be greater than 2

7. PD Index 4, PTR, shall be 0

8. DE Field 7, Transformation Matrix, shall be 0
DE Field 15, Form, shall be 0

9. DE Field 15, Form, shall be 0

10. DE Field 15, Form, shall be either 9 or 0.

11. PD Index 2, Number of Leaders, shall be non-zero.
All of the pointer fields, DENOTE and DE1 through DEN, are required to be present and shall point to valid entities.

* 12. PD Index 5, Font Characteristic, shall be 1, 1001, 1002 or 1003

13. DE Field 9b, Subordinate Status, shall be 01
DE Field 7, Transformation Matrix, shall be 0

14. PD Index 4, ZT, shall be 0.0

15. DE Field 15, Form, shall have the value 7.

16. PD Indices 3 through 2+NC, pointers to parent and children, shall be pointers to planes (Type 108).

17. If 2-D wireframe descriptions are used, Z-coordinates shall be 0.0

* Denotes an IGES Version 4.0 capability

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3.2.4.3. Entity construction. The following entities (in entity number order) have particular meanings when used for NC Manufacturing. The requirements in this section shall be met in all conforming data files and by all translator implementations of the Class IV subset of ANSI Y14.26M.

100 Circular Arc. The circular arc is used to represent circular edges of a part. When using the circular arc to represent a complete circle, the start point and the end point shall be the same and shall be on the positive XT axis (PD indices 3,5,and 7 shall be identical, and PD indices 4,6,and 8 shall be identical).

102 Composite Curve. Composite curves are intended for showing connectivity and continuity among a number of component geometry entities. For example, composite curves are used to represent the profile of a part.

104 Conic Arc. The conic arc is used to represent elliptical, hyperbolic, and parabolic edges of the part. When using the conic arc to represent a full ellipse, the start point and the terminate point shall be the same and shall be on the positive XT axis. (PD indices 9 and 11 shall be 0.0, and PD indices 8 and 10 shall be identical).

110 Line. All lines shall have a non-zero length (The two endpoints of a line shall not be identical).

124 Transformation Matrix. Defining matrices (Forms 0 and 1) are used to position an entity into model space from its definition space. When entities share the same plane of definition, they shall use the same transformation matrix to avoid multiple identical matrices being included in the file. If an entity contains translation information in its PD section, a transformation matrix shall not be used to translate the entity.

126 Rational B-spline Curve. The rational B-spline curve is used to represent free-form edges of a part. It shall not be used to represent linear, circular, or conic edges of a part since more appropriate entities exist for these shapes. Field 16, form, shall be zero.

202 Angular Dimension. This entity shall have a general note, two leaders, and a vertex point. The Z displacement of the vertex point can be attained from any of the subordinates.

206 Diameter Dimension. This entity shall have a general note, at least one leader, and an arc center (may not be defaulted). The Z displacement of the arc center can be attained from the subordinates. The arc center shall be valid. If multiple leaders occur, the first segment of each leader shall be co-linear and opposite in direction.

210 General Label. This entity shall have at least one leader and one general note.

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212 General Note. General notes shall use a font code to minimize the number of text strings in the note. At least one string is required, but the number of different strings shall be minimized. Form numbers plus position information on each string shall be used. The Rotation Angle field shall contain the string angle. Transformation matrices shall not be used for string angles. Null strings are allowed and may be used to pattern a note into one of the standard forms.

216 Linear Dimension. This entity shall have two leaders and at least one general note. The witness lines shall be parallel.

218 Ordinate Dimension. This entity shall have a general note and either a leader or witness line. This entity shall not be used in place of the general label. The leader shall contain only one segment.

222 Radius Dimension. This entity shall have a general note, at least one leader, and an arc center (may not be defaulted). Z displacement of the arc center can be attained from any of its subordinates. Arc center shall be valid. If multiple leaders occur, the first segment of each shall be parallel. If two leaders are used, one and only one shall be Form 4.

404 Drawing. The drawing entity defines the basic engineering drawing format for each sheet. One drawing entity shall exist in the file for each sheet of an engineering drawing.

406 Form 16 Drawing Size. A drawing size property shall be included for each drawing entity in the file. If drawing size is not in the architecture of the system, the pre-processor shall insert one.

406 Form 17 Drawing Units. A drawing units property shall be included for each drawing entity in the file. If drawing units is not in the architecture of the system, the pre-processor shall use the defaults given in Global Parameters 14 and 15.

In general, the simplest entity type shall be used to represent each piece of geometry, and zero size entities shall not be used. For instance, a B-Spline Curve shall not be used to represent a circular arc. A zero length line entity or a zero diameter circle shall not occur in the file.

All 200 series entities shall be flagged as annotation and shall be parallel to the viewing plane, except for crosshatching. All dependent entities with parent 200 entities shall be flagged as annotation and shall be coplanar with their parent. Any annotation entity which points to multiple leaders and multiple witness lines shall order those leaders and witness lines so that leader-1 corresponds logically to witness-1 and leader-2 to witness-2.

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3.2.4.4. File construction.

3.2.4.4.1. Start section. The following information shall be given in the Start Section of the file:

a. Statement of conformance to this application subset and date.

b. All part and drawing identification data required in the drawing title block by DOD-STD-100 and DOD-D-1000.

c. Revision level of the file.

d. Performing organization, date and contract number.

e. Data organization method with contents of each level.

3.2.4.4.2. Global section. Fields in the global section shall be restricted to certain ranges and shall not be defaulted except as noted below for parameters 1, 2, 12 and 24.

Field	Value	Required	
1	,	N	Default to ,
2	;	N	Default to ;
3-11		Y	
12		N	Default to Field 3
13	1.0	Y	
14	1-11	Y	
15		Y	
16		Y	
17		Y	
18-22		Y	
23	4-6	Y	
24	0-7	N	Default to zero

3.2.4.4.3. Directory entry section. See notes for table IV for restrictions placed on the parameters in the directory entry section. In addition, the following capabilities shall be provided and shall be supported for all entities:

a. Line font pattern.
b. The level number field shall be zero or positive except where necessary to maintain the meaning of the referenced entity.
c. Translation matrix.
d. Blank status flag.
e. Subordinate entity switch.
f. Entity use flag.
g. Hierarchy status flag.
h. Line weight number.

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i. Color number.
j. Form number.

3.2.4.4.4. Parameter data section. See notes for table IV for restrictions placed on the parameters in the parameter data section.

3.2.4.5. Mapping of information content to ANSI Y14.26M subset entities. NC Manufacturing geometry shall be mapped into ANSI Y14.26M geometry entities and linked together as necessary with composite curve entities. Text shall be represented by the general note entities and shall not be represented as geometry. Annotation including dimensions, labels and centerlines shall be represented by their named ANSI Y14.26M entity and shall not be represented as geometry. Line width, color, and line font information shall be represented by the appropriate global and directory entry parameters. Level attributes shall be represented by the appropriate directory entry parameter.

3.2.4.6. Data accuracy requirements. All data transformations shall maintain an accuracy of 1.0×10^{-6} units on all parametric and coordinate values and all measurable dimensions.

3.2.4.7. User conventions and data organization. A minimum complexity drawing and view entity combination shall be used to assure a part model being created on all destination systems. A drawing size property shall be used to define drawing limits. As specified in the contract or other form of agreement, data shall be organized as one drawing per file with multiple sheets permitted, or shall be restricted to one sheet per file.

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4. QUALITY ASSURANCE PROVISIONS.

4.1. Responsibility for inspection. Unless otherwise specified in the contract or other form of agreement, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or other form of agreement, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.2. Responsibility for compliance. All items shall meet all requirements of section 3. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.3. Inspection procedures.

4.3.1. Technical illustration subset. Start, global, directory entry, parameter data and terminate sections shall each be analyzed for conformance to ANSI Y14.26M with an appropriate software utility. The start section shall be displayed and checked visually with requirements of 3.2.1.3.1. The global section shall be displayed and compared against the requirements of 3.2.1.3.2. Entities in the directory entry section shall be checked against table I by an appropriate software utility. Ranges of parameter values shall be compared against requirements of table I.

4.3.2. Engineering drawing subset. Start, global, directory entry, parameter data and terminate sections shall each be analyzed for conformance to ANSI Y14.26M with an appropriate software utility. The start section shall be displayed and checked visually with requirements of 3.2.2.4.1. The global section shall be compared against the requirements of 3.2.2.4.2. Entities in the directory entry section shall be checked against table II by an appropriate software utility. Ranges of parameter values shall be compared against requirements of table II.

4.3.3. Electrical/electronic applications subset. Start, global, directory entry, parameter data and terminate sections shall each be analyzed for conformance to ANSI Y14.26M with an appropriate software utility. The start section shall be displayed and checked visually with requirements of 3.2.3.5.1. The global section shall be compared against the requirements of 3.2.3.5.2. Entities in the directory entry shall be checked against table III

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by an appropriate software utility. Ranges of parameter values shall be compared against requirements of table III.

4.3.4. Geometry for NC Manufacturing subset. Start, global, directory entry, parameter data and terminate sections shall each be analyzed for conformance to ANSI Y14.26M with an appropriate software utility. The start section shall be displayed and checked visually with requirements of 3.2.4.4.1. The global section shall be compared against the requirements of 3.2.4.4.2. Entities in the directory entry section shall be checked against table IV by an appropriate software utility. Ranges of parameter values shall be compared against requirements of table IV.

5. PACKAGING.

Packaging of product data files for delivery shall be in accordance with the requirements of MIL-STD-1840.

6. NOTES.

6.1. Intended use. This specification is designed to be incorporated into a contract to define the technical requirements to be met when it is desired to purchase product definition data or product data in digital form. IGES is a specification for representing digital product definition data in a neutral, public domain format. IGES provides a neutral format for the representation and transfer of vector graphics data used for illustration purposes among CAD systems and application programs. Information is transferred by entities that represent geometry, annotation, attribute and logical relationships of the product model. IGES files are composed of five sections; Start, Global, Directory Entry, Parameter Data, and Terminate. An additional section is required for the Compressed physical format.

This specification defines the technical requirements for the exchange of digital product data in specific application subsets of ANSI Y14.26M. The essential content and general requirements of application subsets are given by this specification and specific application subsets are identified. The list of subsets covered is not all inclusive and can be expected to grow as more application subsets are defined and developed. In particular the definition and use of the following specific application areas are defined:

a. Technical Illustrations - Class I subset
b. Engineering Drawings - Class II subset
c. Electrical/Electronic Applications - Class III subset
d. Geometry for NC Manufacturing - Class IV subset

The number of defined subsets is expected to grow as product data exchanges become commonplace, as technical groups identify and define specific

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application areas, and as these application subset requirements can be validated.

Listed below are other application areas being characterized and defined.

a. Process Plant Flowsheets
b. 3-D Piping and Tubing
c. 3-D Mechanical Design/Drawing
d. Finite Element Modeling
e. Architecture, Engineering and Construction
f. Mechanical Solid Modeling

In some cases, work is well underway to define ANSI Y14.26M subsets for these application areas. It is expected that future revisions of this military specification will include some of these application subsets after they have been approved by the appropriate IGES committees.

It is believed that the subsets in this document are not specific to DOD needs. It is therefore intended that with industry collaboration these subsets be incorporated within (or appended to) ANSI Y14.26M so that the need to separately identify them in a military specification is eventually eliminated.

Assuring completeness of information exchange with this subset concept relies on a careful encoding of the information into the IGES entities. This specification is a first attempt at this encoding. But a more rigorous method is now under development. Called an Application Protocol, the new work involves a formal information model with the rigorouos mapping to the IGES entities. It is the intent of this specification to evolve in the direction of application protocols to ensure quality data exchanges.

6.2. Ordering data. The contract or other form of agreement should specify the following:

a. Title, number, and date of this specification.

b. The type of product data being procured as class I, class II, class III, or class IV. (See 3.1)

c. The form of the ANSI Y14.26M data file as ASCII or Compressed ASCII. (See 3.1.3.)

d. The physical media to be used if not magnetic tape. (See 3.1.4.)

e. For class II, engineering drawings, specify:

1. One drawing per file with multiple sheets per file permitted. (See 3.2.2.7.)

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2. One sheet per file. (See 3.2.2.7.)

6.3. File size and efficiency considerations. IGES files are often quite large, and implementors are urged to make use of the most efficient entity constructs. This specification calls for the use of entity constructs in the form of the ANSI Y14.26M standard as extended by IGES Version 4.0 for the dotted line font (DE Field 4=5) and the intercharacter spacing property (Type 406 Form 18).

Several of the entity constructs included in the class subsets of this specification were selected to keep the file sizes to acceptable levels. For instance, the use of subfigures greatly reduces file size where details are repeated; however, more verbose entity construct may be essential, such as stroking text characters to obtain special appearance like proportional spacing or using multiple line segments to provide a dotted line font.

Use of more efficient IGES Version 4.0 constructs like the dotted line font or the intercharacter spacing property for proportional text spacing is encouraged to reduce file sizes. If the intercharacter spacing property is used, then all IGES Version 4.0 character fonts may be used.

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6.4. Summary of start section requirements. The start section of an IGES data file is used to contain human-readable notes to aid in interpreting the data in the file. This specification makes use of such capability, and requirements are stated in appropriate places in the class subset descriptions. Specifically, the following paragraphs reference notes to be placed in the start section:

3.1	3.2.2.4.1
3.2.1.3.1	3.2.3.5.1
3.2.4.4.1	

The format or appearance of the notes is not specified and is left to the discretion of the implementor. However, for explanation a sample start section is shown below.

THIS FILE CONTAINS ILLUSTRATIONS FOR A	S00000001
TECHNICAL DOCUMENT IN MIL-D-28000	S00000002
CLASS I FORM - 22 DECEMBER 1987	S00000003
	S00000004
FIGURE NUMBER 12-B	S00000005
	S00000006

6.5. Additional processing conditions. Certain additional IGES file processing practices are preferred, but are not mandatory. Implementors should be aware of the following Recommended Practices (RP), from the IGES Recommended Practices Guide, which further specify preferred processing algorithms. These include:

RP 1: Delimiter
RP 2: Witness Line Suppression
RP 4: Transformation Matrix Processing
RP 5: System ID Parameter
RP 7: Maximum Coordinate Value
RP 8: Independent Witness Lines
RP 12: Flag Note Restrictions
RP 15: Zero Radius Arcs
RP 16: Translation Vector
RP 17: Model Space Scale
RP 19: Independent and Dependent Processing
RP 20: Back Pointers in View Associativity
RP 21: Comments in PD Records
RP 22: Bounded Planes
RP 24: Representation of Linear Strings

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D- 28000/AMEND. 1	DIGITAL REPRESENTATION FOR COMMUNICATION OF PRODUCT DATA: IGES

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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